

Reply Notes for Emilien Ouvrard

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Direct answers to your questions

Your setup and your diagnostics are sensible. You are asking the right questions.

You wrote: “I am not yet using Galerkin/LSPG, I am testing whether GPR can learn $k \mapsto q(k)$.”

That is a good first test. And since you already state this clearly, I will only make the distinction explicit so we are fully aligned.

In your current test, you learn a direct map

$$k \mapsto \mathbf{q}_N(k).$$

with reconstruction

$$e_h(k) \approx \mathbf{V}_N \mathbf{q}_N(k).$$

In the closure setup, we split

$$\mathbf{V}_N = [\mathbf{V}_p \ \mathbf{V}_s], \quad \mathbf{q}_N = \begin{bmatrix} \mathbf{q}_p \\ \mathbf{q}_s \end{bmatrix},$$

where $\mathbf{q}_p$ are the retained/primary coefficients and $\mathbf{q}_s$ are the secondary coefficients.

Then the learned relation is typically

$$\mathbf{q}_s \approx g(\mathbf{q}_p),$$

and $\mathbf{q}_p$ comes from the reduced projection solve (Galerkin or LSPG), not from direct regression on k . In intrusive form:

$$\mathbf{q}_p(k) = \arg \min_{\mathbf{q}_p} \|R(\mathbf{V}_p \mathbf{q}_p + \mathbf{V}_s g(\mathbf{q}_p); k)\|^2.$$

So yes, both approaches use GPR, but they are different regression problems. Because of that, if direct POD-GPR on $k \mapsto \mathbf{q}_N$ struggles, it does *not* automatically mean PROM-GPR will fail.

Your Question 1: “Are you surprised by irregularity in modes 12–19? Should these be smoother?”

I am not surprised.

I will refer to q_i as coefficients (not modes). Higher-index POD coefficients usually carry lower-energy and more oscillatory/localized content. They are often less regular and more sensitive to:

- sampling density in parameter/frequency,

- numerical conditioning near resonances.

To be very direct: in my Burgers case, if I take one test trajectory and plot coefficients q_1 – q_4 , q_{21} – q_{24} , and q_{101} – q_{104} , the first coefficients are clean and structured while higher-index coefficients are clearly rougher (see Figure A). Then, when I compare true vs predicted on the same trajectory (Figure B), the larger errors concentrate in the highest-index coefficients. So this roughness pattern is something we do see in practice; it is not, by itself, a red flag.

The two checks I would prioritize first are:

- POD singular-value decay and cumulative energy: if those irregular coefficients belong to very low-energy directions, their impact on the reconstructed field is usually limited.
- Plot $\Re(q_i)$ and $\Im(q_i)$ directly, not only $|q_i|$: for complex coefficients, the modulus can look artificially jagged when amplitude is small and phase rotates.

Your Question 2: “What was the shape of your data? Is GPR still suitable?”

This is a key point.

In our workflow, the effective training set is usually built *per snapshot state*, not only per parameter point. So we often have many more training rows than “number of parameter samples”.

For example, in one of our reference runs (different PDE, but same data-organization idea), we had:

- $\mathbf{q}_p \in \mathbb{R}^{20 \times 4509}$,
- $\mathbf{q}_s \in \mathbb{R}^{131 \times 4509}$,
- test trajectories with 501 time samples.

So the model sees thousands of state samples, not only a few dozen parameter points.

In your current direct-frequency map, with around 90 training frequencies and one scalar input (k), GPR can still be a good choice, but it is a tougher regime for learning all coefficients at once.

Linear PROM performance: how I would interpret it

Before deciding on nonlinear closure, I would still check a linear PROM baseline very carefully.

There are two practical cases:

- If linear PROM is already accurate with relatively few retained coefficients, there may be no urgent need for PROM-GPR.
- If linear PROM needs many retained coefficients to reach the target accuracy, that is exactly the regime where PROM-GPR can be useful to reduce online dimension.

So if direct POD-GPR ($k \mapsto \mathbf{q}_N$) is weak, I would not stop there. I would still:

- first secure accuracy with linear PROM (even if many coefficients are needed),
- then use PROM-GPR to compress/recover accuracy with a smaller effective online dimension.

In other words, there are problems where an intrusive, physics-based PROM is necessary because a purely non-intrusive map (for example direct POD-GPR or POD-DNN without projection/physics in the loop) is difficult to learn robustly.

Your Question 3: “What method could handle rough data? DNN maybe?”

With your current data volume, I would still start by pushing GPR further before moving to DNNs. Concretely, I would test:

- one GP per output coefficient (instead of one shared multi-output fit), so each coefficient can have its own kernel and its own shape/lengthscale parameters;
- kernel families beyond a single smooth stationary choice (for example Matérn vs RBF vs Rational Quadratic, and combinations), because wave-like responses can require different regularity assumptions coefficient-by-coefficient;
- sparse/inducing-point GP variants if you later scale to much larger datasets (with 90 samples this is usually not required, but it can matter once your dataset grows).

For anisotropy/ARD: the key point is that ARD assigns one lengthscale per *input feature*. If input is only k , there is only one input-direction lengthscale. In a PROM-GPR closure setting, input becomes $\mathbf{q}_p$, so input dimension is n_p and ARD becomes much more meaningful (one lengthscale per component of $\mathbf{q}_p$). You still can have different shape parameters across coefficients by fitting independent GP models per output coefficient.

About error interpretation: a relative error of around 15% on a high-index coefficient does not necessarily imply a 15% solution error. Two reasons:

- high-index coefficients often have lower energy, so their contribution to the reconstructed field is smaller;
- for oscillatory/wave-like signals, relative coefficient error can be inflated by phase shift or local amplitude mismatch, even when the global field discrepancy is moderate.

So I would report both:

- coefficient-level errors (relative and absolute),
- and solution-level metrics (field norm errors, plus phase/amplitude-aware metrics when relevant; Wasserstein-type metrics can also be informative in some settings).

Optional supporting figures

A. Typical low-vs-high mode behavior

B. True vs predicted overlays at test point (4.56, 0.019)

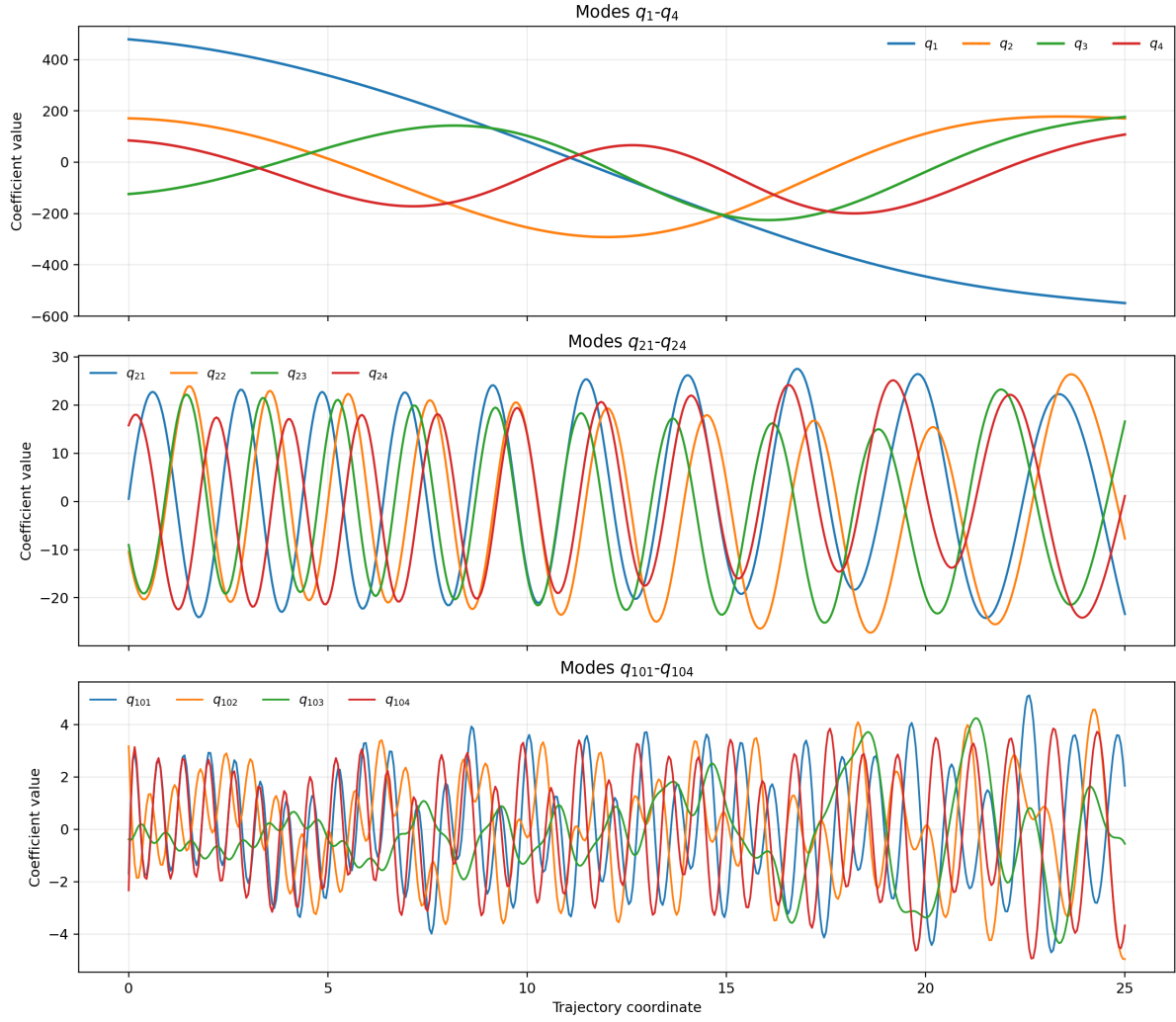


Figure 1: Illustrative coefficient trajectories from one internal test trajectory using the three requested blocks: q_1 - q_4 , q_{21} - q_{24} , and q_{101} - q_{104} .

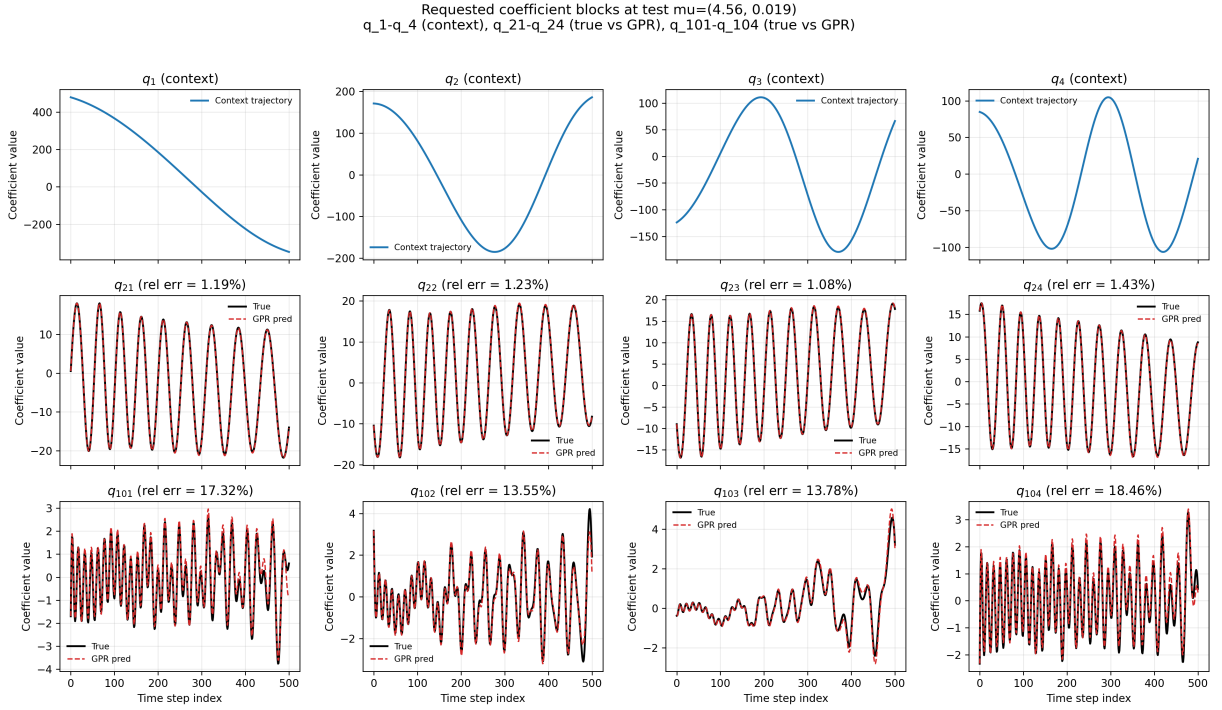


Figure 2: Coefficient overlays with the requested blocks: q_1 - q_4 , q_{21} - q_{24} , and q_{101} - q_{104} . Here q_1 - q_4 are shown as context coefficients, while q_{21} - q_{24} and q_{101} - q_{104} are secondary coefficients with true-vs-GPR comparison.